From topology to combinatorics via computability theory

Damir D. Dzhafarov
University of Connecticut
`www.damirdzhafarov.com`

Summer School on "Reverse Mathematics: New Paradigms"
ESI · Vienna · July 2025

Recap

GS. Every infinite CSC space has an infinite subspace that is indiscrete, has the initial segment topology, has the final segment topology, is discrete, or has the cofinite topology.

RT^2_2: Every coloring $c : [\mathbb{N}]^2 \to 2$ has an infinite homogeneous set.

CAC: Every partial order $\leq_P$ on $\mathbb{N}$ has an infinite chain or antichain.

ADS: Every linear order $\leq_L$ on $\mathbb{N}$ has an infinite ascending sequence or descending sequence.

Lecture 2

Proving GS in ACA_0 (formalizing original proof)

Let $\langle \mathbb{N}, \mathcal{U}, k\rangle$ be a CSC space, with $\mathcal{U} = \langle U_n : n \in \mathbb{N}\rangle$.

By arithmetical comprehension, form $\mathrm{cl} = \{\langle x, y\rangle : (\forall i)[x \in U_n \to y \in U_n]\}$.

(Write $x \in \mathrm{cl}(y)$ if $\langle x, y\rangle \in \mathrm{cl}$.)

Using cl, we define the equivalence relation $\sim$ and the partial order $\trianglelefteq$.

(This can be done even in RCA_0.)

In the T_0 case we apply CAC to $\trianglelefteq$. (Provable in ACA_0.)

In the case of an infinite $\trianglelefteq$-chain we apply ADS. (Provable in ACA_0.)

In the case of an infinite $\trianglelefteq$-antichain, either some subspace has the cofinite topology or we carry out the inductive construction of a discrete subspace.

(Required asking if certain sets had infinite complements: Π^0_2.)

Variations of GS

The following statements are defined in RCA_0.

GS. Every infinite CSC space has an infinite subspace that is indiscrete, has the initial segment topology, has the final segment topology, is discrete, or has the cofinite topology.

wGS. Every infinite CSC space has an infinite subspace that is indiscrete, has the initial segment topology, has the final segment topology, or is T_1.

GST_1. GS restricted to T_1 CSC spaces.

GS^{cl}. GS restricted to CSC spaces for which the closure operator exists.

wGS^{cl}. wGS restricted to CSC spaces for which the closure operator exists.

Relationships to the proof of G–S

Principle	Measures the strength of
GS	Full Ginsburg–Sands theorem (for c.s.c. spaces).
$\mathsf{GS}^{\mathrm{cl}}$	Full theorem, minus the complexity of building $\sim$ and $\trianglelefteq$.
wGS	Full theorem, minus the complexity of handling the T_1 case.
$\mathsf{wGS}^{\mathrm{cl}}$	Full theorem, minus the complexity of building $\sim$ and $\trianglelefteq$, and of handling the T_1 case.
GST_1	The construction handling the T_1 case.

Relationships to the proof of G–S

Principle	Measures the strength of
GS	Full Ginsburg–Sands theorem (for c.s.c. spaces).
$\mathsf{GS}^{\mathrm{cl}}$	Full theorem, minus the complexity of building $\sim$ and $\trianglelefteq$.
wGS	Full theorem, minus the complexity of handling the T_1 case.
$\mathsf{wGS}^{\mathrm{cl}}$	Full theorem, minus the complexity of building $\sim$ and $\trianglelefteq$, and of handling the T_1 case.
GST_1	The construction handling the T_1 case.

The strength of $\mathsf{wGS}^{\mathrm{cl}}$

Theorem. Over RCA_0, $\mathsf{wGS}^{\mathrm{cl}} \leftrightarrow \mathsf{CAC}$.

That $\mathsf{CAC} \to \mathsf{wGS}^{\mathrm{cl}}$ is what we saw in the proof of GS.

Let $(P, \leq_P)$ be an infinite partial order. Assume there is no infinite $\leq_P$ antichain.

For $p \in P$, define $V_p = \{q \in P : p \leq_P q\}$.

Let $\langle P, \mathcal{U}, k \rangle$ be the CSC space generated by the V_p.

Claim. $p \in \mathrm{cl}(q)$ if and only if $p \leq_P q$.

- If $p \in \mathrm{cl}(q)$ then $q \in V_p$ so $p \leq_P q$.
- If $p \notin \mathrm{cl}(q)$ then there is some $U \in \mathcal{U}$ containing p but not q. So for some r $p \in V_r$ and $q \notin V_r$. This means $r \leq_P p$ but $r \not\leq_P q$, hence $p \not\leq_p q$.

The strength of $\mathsf{wGS}^{\mathsf{cl}}$

It follows that the closure operator on $\langle P, \mathcal{U}, k\rangle$ exists.

We can thus apply $\mathsf{wGS}^{\mathsf{cl}}$ to get an infinite subspace Y of P.

- By our opening assumption, Y is not an $\leq_P$-antichain.
- Fix $p <_p q$ in Y. Then $p \in \mathrm{cl}(q)$ and $p \notin U_q$.
- Hence, Y is neither T_1 nor indiscrete.
- If Y has the initial segment topology, fix a witnessing bijection $h : \mathbb{N} \to Y$.
- We claim $n < m \longrightarrow h(m) <_p h(n)$. (So Y is $\leq_P$-descending.)
- Fix $n < m$ and $U \in \mathcal{U}$ with $h(m) \in U$. Fix j so that $U \cap Y = \{h(i) : i \leq j\}$.
- Then $n < m \leq j$, so also $h(n) \in U$.
- We conclude that $h(m) \in \mathrm{cl}(h(n))$, and hence $h(m) <_P h(n)$ as wanted.

Relationships to the proof of G–S

Principle	Measures the strength of
GS	Full Ginsburg–Sands theorem (for c.s.c. spaces).
$\mathrm{GS}^{\mathrm{cl}}$	Full theorem, minus the complexity of building $\sim$ and $\trianglelefteq$.
wGS	Full theorem, minus the complexity of handling the T_1 case.
$\mathrm{wGS}^{\mathrm{cl}}$	Full theorem, minus the complexity of building $\sim$ and $\trianglelefteq$, and of handling the T_1 case.
GST_1	The construction handling the T_1 case.

Relationships to the proof of G–S

Principle	Measures the strength of
GS	Full Ginsburg–Sands theorem (for c.s.c. spaces).
$\mathsf{GS}^{\mathrm{cl}}$	Full theorem, minus the complexity of building $\sim$ and $\trianglelefteq$.
wGS	Full theorem, minus the complexity of handling the T_1 case.
$\mathsf{wGS}^{\mathrm{cl}}$	Full theorem, minus the complexity of building $\sim$ and $\trianglelefteq$, and of handling the T_1 case.
GST_1	The construction handling the T_1 case.

The strength of $\mathsf{GS}^{\mathrm{cl}}$

Theorem. Over RCA_0, $\mathsf{GS}^{\mathrm{cl}} \leftrightarrow \mathsf{GST}_1$.

($\mathsf{GST}_1 \to \mathsf{GS}^{\mathrm{cl}}$) Let X be a CSC space whose closure operator exists.

- Using the closure operator we can define $\sim$ and $\trianglelefteq$ from the classic proof.
- We will see that $\mathsf{GST}_1 \to \mathsf{RT}^2_2$, hence also CAC and ADS. (Soon!)
- Hence, if necessary we can reduce to the T_1 case and apply GST_1.

($\mathsf{GS}^{\mathrm{cl}} \to \mathsf{GST}_1$) Let X be a T_1 CSC space.

- The closure operator is trivial: $x \in \mathrm{cl}(y) \leftrightarrow x = y$.
- Apply $\mathsf{GS}^{\mathrm{cl}}$ to get an infinite subspace Y. Since X is T_1, this subspace cannot be indiscrete, nor have the initial or final segment topology.
- It follows that Y is discrete or has the cofinite topology.

Relationships to the proof of G–S

Principle	Measures the strength of
GS	Full Ginsburg–Sands theorem (for c.s.c. spaces).
$\mathrm{GS}^{\mathrm{cl}}$	Full theorem, minus the complexity of building $\sim$ and $\trianglelefteq$.
wGS	Full theorem, minus the complexity of handling the T_1 case.
$\mathrm{wGS}^{\mathrm{cl}}$	Full theorem, minus the complexity of building $\sim$ and $\trianglelefteq$, and of handling the T_1 case.
GST_1	The construction handling the T_1 case.

Relationships to the proof of G–S

Principle	Measures the strength of
GS	Full Ginsburg–Sands theorem (for c.s.c. spaces).
$\mathrm{GS}^{\mathrm{cl}}$	Full theorem, minus the complexity of building $\sim$ and $\trianglelefteq$.
wGS	Full theorem, minus the complexity of handling the T_1 case.
$\mathrm{wGS}^{\mathrm{cl}}$	Full theorem, minus the complexity of building $\sim$ and $\trianglelefteq$, and of handling the T_1 case.
GST_1	The construction handling the T_1 case.

Existence of the closure operator

Theorem. Over RCA_0, the statement that, for every CSC space the closure operator exists, is equivalent to ACA_0.

Assume the statement and fix $f : \mathbb{N} \to \mathbb{N}$. We prove $\mathrm{ran}(f)$ exists.

Define $V_{\langle x,s\rangle} = \begin{cases} \{2x, 2x+1\} & \text{if } (\forall y < s)[f(y) \neq x], \\ \{2x\} & \text{otherwise.} \end{cases}$

In RCA_0, we can generate a CSC space $X = \langle \mathbb{N}, \mathcal{U}, k\rangle$ from these sets.

(We add $\emptyset$, X, and define k using Δ^0_1-CA.)

Now given the closure operator cl on X, we have that

$$x \in \mathrm{ran}(f) \leftrightarrow \{2x\} \in \mathcal{U} \leftrightarrow 2x \notin \mathrm{cl}(2x+1).$$

The bounding schema

Let Γ be a class of formulas (e.g., Σ^0_n, Π^0_n).

The Γ-**bounding schema** contains, for each formula $\varphi(x, y) \in \Gamma$, the axiom

$$(\forall n)[(\forall x < n)(\exists y)\varphi(x, y) \to (\exists b)(\forall x < n)(\exists y < b)\varphi(x, y)]. \qquad (\mathsf{B}\Gamma)$$

We have the **Kirby-Paris hierarchy**:

$$\mathsf{I}\Sigma^0_1 < \mathsf{B}\Sigma^0_2 \leftrightarrow \mathsf{B}\Pi^0_1 < I\Sigma^0_2 < \mathsf{B}\Sigma^0_3 \leftrightarrow \mathsf{B}\Pi^0_2 < \mathsf{I}\Sigma^0_3 < \cdots.$$

(RCA$_0$ does not prove $\mathsf{B}\Sigma^0_2$.)

Theorem (Hirst). Over RCA$_0$, $\mathsf{B}\Sigma^0_2 \leftrightarrow (\forall k)\mathsf{RT}^1_k$.

(In some models, the number of colors k may be nonstandard.)

The bounding schema

Theorem (Hirschfeldt and Shore). Over RCA_0, $\mathsf{CAC} \to \mathsf{B}\Sigma^0_2$.

We argue in RCA_0 and prove $(\forall k)\mathsf{RT}^1_k$. Fix $k \in \mathbb{N}$ and $c : \mathbb{N} \to k$.

Define $\leq_P$ as follows: let $x <_P y$ if $x < y$ and $c(x) = c(y)$.

(Transitivity: if $x <_P y <_P z$ then $x < y < z$ and $c(x) = c(y) = c(z)$, so $x <_P z$.)

Apply CAC to get an infinite set $X = \{x_0 < x_1 < \cdots\}$.

- If X is a $\leq_P$-chain then $x_0 <_P x_1 <_P < \cdots$ so $c(x_0) = c(x_1) = \cdots$.
  Hence, X is an infinite homogeneous set for c.
- So suppose X is a $\leq_P$-antichain. Then $c(x_i) \neq c(x_j)$ for all $i \neq j$.
  But then $c \restriction \{x_0, \ldots, x_k\}$ is a finite injection of $k + 1$ into k.

($\mathsf{I}\Sigma^0_1$ suffices to prove this is impossible.)

Classification of GS

Theorem. Over RCA_0, GS and wGS are both equivalent to ACA_0.

We already have $\mathsf{ACA}_0 \to \mathsf{GS} \to \mathsf{wGS}$. So: remains to show $\mathsf{wGS} \to \mathsf{ACA}_0$.

In RCA_0, fix $f : \mathbb{N} \to \mathbb{N}$. Write $\mathrm{ran}(f) \restriction w\,[x] = \{z < w : (\exists y \leq x)[f(y) = z]\}$.

For all x, s, define $V_{x,s} = \{x\} \cup \{w < x : \mathrm{ran}(f) \restriction w\,[x] = \mathrm{ran}(f) \restriction w\,[x+s]\}$.
Let $\langle \mathbb{N}, \mathcal{U}, k \rangle$ be the CSC space generated by the $V_{x,s}$.

Claim 1. For every w there are cofinitely many x such that $x \in \mathrm{cl}(w)$.

- Fix $x > w$ so that $\mathrm{ran}(f) \restriction w\,[x] = \mathrm{ran}(f) \restriction w$. ($\mathsf{B}\Sigma^0_2$. But $\mathsf{wGS} \to \mathsf{wGS}^{\mathrm{cl}}$.)
- We have $x \in V_{x,s}$ for all s, but also $w \in V_{x,s}$ since $w < x$ and by choice of x.
- Suppose $x \in V_{z,s}$ for some $z > x$. Then $\mathrm{ran}(f) \restriction x\,[z] = \mathrm{ran}(f) \restriction x\,[z+s]$. But $w < x$ so also $\mathrm{ran}(f) \restriction w\,[z] = \mathrm{ran}(f) \restriction w\,[z+s]$, hence $w \in V_{z,s}$.

Classification of GS

Claim 1. For every w there are cofinitely many x such that $x \in \mathrm{cl}(w)$.

Apply wGS to $\langle \mathbb{N}, \mathcal{U}, k \rangle$ to get an infinite subspace $Y = \{y_0 < y_1 < \cdots\}$.

By Claim 1, Y cannot be T_1 or have the final segment topology.
Also, if $x < y$ then $V_{x,0}$ contains x but not y, so Y cannot be indiscrete.

So Y has the initial segment topology, say with witnessing bijection $h : \mathbb{N} \to Y$.
(So for $U \in \mathcal{U}$ we have $U \cap Y = \{h(i) : i \leq j\}$ for some j.)

Claim 2. $h(n) = y_n$ for all n.

- Suppose there exist $n < m$ such that $h(n) > h(m)$. Choose j so that $V_{h(m),0} \cap Y = \{h(i) : i \leq j\}$. Then $h(n) \in V_{h(m)}$ since $n < m$, contradiction.
- So $n < m \to h(n) < h(m)$. Since h is bijective, $h(n) = y_n$. $(\mathrm{I}\Sigma^0_1)$

Classification of GS

Claim 2. $h(n) = y_n$ for all n.

It follows that every basic open set in Y is closed downward under $<$.

(More precisely: if $U \in \mathcal{U}$ then $(x, z \in Y \wedge x < z \wedge z \in U) \to x \in U$.)

We now prove $\mathrm{ran}(f)$ exists. Fix w, and choose $x, z \in Y$ so that $w < x < z$.

Claim 3. $\mathrm{ran}(f) \restriction w = \mathrm{ran}(f) \restriction w\,[z]$.

- Assume not. Fix s so that $\mathrm{ran}(f) \restriction w\,[z] \neq \mathrm{ran}(f) \restriction w\,[z+s]$.
- In particular, $\mathrm{ran}(f) \restriction x\,[z] \neq \mathrm{ran}(f) \restriction x\,[z+s]$ since $w < x$.
- But then $V_{z,s}$ contains z but not x, even though $x < z$ and $x, z \in Y$.

Stable colorings

A coloring $c : [\mathbb{N}]^2 \to 2$ is **stable** if $(\forall x)(\exists i < 2)(\forall^\infty y)[c(x, y) = i]$.

(That is, $\lim_y c(x, y) = i$ and i is the **limit color** of x.)

RT^2_2: Every coloring $c : [\mathbb{N}]^2 \to 2$ has an infinite homogeneous set.

SRT^2_2: Every stable coloring $c : [\mathbb{N}]^2 \to 2$ has an infinite homogeneous set.

A set $L \subseteq \mathbb{N}$ is **limit homogeneous** for c if every $x \in L$ has the same limit color.

- Every such L can be **thinned** to an infinite homogeneous set H. If c is computable, then this thinning can be done computably. (So $H \leq_T L$.)
- Chong, Lempp, and Yang showed $\mathsf{SRT}^2_2 \leftrightarrow \mathsf{D}^2_2$ over RCA_0.

D^2_2: Every stable coloring $c : [\mathbb{N}]^2 \to 2$ has an infinite limit homogeneous set.

Pairs to singletons

A stable $c : [\mathbb{N}]^2 \to 2$ induces a coloring $d : \mathbb{N} \to 2$ by $d(x) = \lim_y c(x, y)$.

- Every infinite homogeneous set for d is a limit homogeneous set for c.
- d is Δ^0_2-definable from c. (So RCA_0 cannot in general prove that d exists.)

Conversely, given a Δ^0_2-definable $d : \mathbb{N} \to 2$ there exists a Δ^0_1-definable stable $c : [\mathbb{N}]^2 \to 2$ such that $d(x) = \lim_y c(x, y)$.

(Shoenfield's limit lemma, formalized in RCA_0.)

Thus, D^2_2 can be thought of as RT^1_2 "one jump up".

In computability theory we can also think of this as follows:

given a stable $c : [\omega]^2 \to 2$, what are the degrees of its infinite homogeneous sets?	$\leftrightarrow$	given a Δ^0_2 set A, what are the degrees of the infinite subsets of A and $\overline{A}$?

Cohesiveness

Given a sequence $\vec{R} = \langle R_n : n \in \mathbb{N} \rangle$ of sets, a set X is **cohesive** for $\vec{R}$ (or **$\vec{R}$-cohesive**) if $(\forall n)[\, |X \cap R_n| < \infty \vee |X \cap \overline{R_n}| < \infty]$.

(Equivalently $(\forall n)[X \subseteq^* R_n \vee X \subseteq^* \overline{R_n}]$.)

COH: Every sequence $\vec{R} = \langle R_n : n \in \mathbb{N} \rangle$ of sets has an infinite $\vec{R}$-cohesive set.

- Either R_0 or $\overline{R}_0$ is infinite. Say R_0 and fix x_0 in it.
- Now either $R_0 \cap R_1$ or $R_0 \cap \overline{R_1}$ is infinite. Say $R_0 \cap \overline{R_1}$ and fix $x_1 > x_0$ in it.
- Continuing, obtain $\{x_0, x_1, \ldots\}$ such that $x_n, x_{n+1}, \ldots$ are all in R_n or all in $\overline{R_n}$.

By identifying each R_n with its characteristic function, COH can be restated as: For every sequence $\langle c_n : n \in \mathbb{N} \rangle$ of colorings $\mathbb{N} \to 2$ there exists an infinite set X which is homogeneous for each c_n modulo a finite set.

Thus, COH can be thought of as a sequential version of RT^1_2 "with finite errors".

Cholak–Jockusch–Slaman decomposition

Theorem (Cholak, Jockusch, and Slaman)
Over RCA_0, $\mathsf{RT}^2_2 \leftrightarrow \mathsf{SRT}^2_2 + \mathsf{COH}$.

Fix $c : [\mathbb{N}]^2 \to 2$ and define $\vec{R} = \{R_x : x \in \mathbb{N}\}$ by $R_x = \{y > x : c(x, y) = 0\}$.

Apply COH to get an infinite $\vec{R}$-cohesive set $X = \{x_0 < x_1 < \cdots\}$.

- If $X \subseteq^* R_x$ then $c(x, y) = 0$ for almost all $y \in X$. (So $\lim_m c(x, x_m) = 0$.)
- If $X \subseteq^* \overline{R_x}$ then $c(x, y) = 1$ for almost all $y \in X$. (So $\lim_m c(x, x_m) = 1$.)

Define $d : [\mathbb{N}]^2 \to \mathbb{N}$ by $d(n, m) = c(x_n, x_m)$. (So d is stable.)

Apply SRT^2_2 to get an infinite homogeneous set H for d.

Then $\{x_n : n \in H\}$ is homogeneous for c.

Cholak–Jockusch–Slaman decomposition

In the other direction, it is clear that $\mathsf{RT}^2_2 \to \mathsf{SRT}^2_2$.

We prove that actually $\mathsf{ADS} \to \mathsf{COH}$.

(Desult due to Hirschfeldt & Shore. The following proof is due to Marcone.)

Given $\vec{R} = \langle R_n : n \in \mathbb{N} \rangle$ define $\vec{S} = \langle S_i : i \in \mathbb{N} \rangle$ by $S_i(n) = R_n(i)$.

Case 1: some S_i appears infinitely often in $\vec{S}$.

(If the R_n are the columns of a matrix, then the S_i are the rows.)

- Let $X = \{j : (\forall n \leq j)[S_j(n) = S_i(n)]\}$. So X is $\vec{R}$-cohesive.

Case 2: else. In RCA_0 we can define a linear order on $\mathbb{N}$ by $i \preceq j$ if $S_i <_{\text{lex}} S_j$.

- Apply ADS to find, e.g., an infinite ascending sequence $i_0 \preceq i_1 \preceq \cdots$.
- For all n we have that $\lim_m S_{i_m}(n) = \lim_m R_n(i_m)$ exists. So X is $\vec{R}$-cohesive.